

Persistent Cognitive Architectures for Human-AI Co-Reasoning: Beyond Retrieval to Structural Intelligence

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Abstract

While retrieval-augmented generation (RAG) enables large language models to access external knowledge, it remains fundamentally reactive and stateless. Current approaches to human-AI collaboration either treat AI as an advanced search tool or propose theoretical frameworks for symbiotic intelligence without demonstrating functional implementations. We present **Symbiose**, a cognitive architecture that combines graph-based causal reasoning, temporal pattern recognition, and proactive autonomy to enable persistent, evolving collaboration between humans and AI.

Components are integrated through **Quantum Leap (QL)**, a persistent cognitive orchestration layer responsible for coordinating memory consolidation, graph-vector fusion, causal reasoning, preference learning, and proactive triggers. Critically, every response is tagged via an **L0-L3 epistemic framework** distinguishing canonical facts (L0) from derived inferences (L1), document-supported claims (L2), and LLM-generated hypotheses (L3), enabling epistemic honesty not achievable in standard LLM deployments without explicit provenance tracking.

Through deployment in cosmological theory development (Energy-Flow Cosmology), we demonstrate emergent capabilities not achievable in stateless architectures without external persistent structure: cross-session identity linking, hierarchical memory consolidation, proactive suggestion generation based on learned preferences and temporal patterns, and explicit provenance tracking for every claim.

Keywords: Human-AI collaboration, cognitive architecture, knowledge graphs, causal reasoning, persistent memory, proactive autonomy

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1 Introduction

1.1 The Limits of Stateless AI

Large language models (LLMs) have demonstrated remarkable capabilities in natural language understanding, reasoning, and generation. However, their deployment in research and knowledge work reveals a fundamental limitation: **statelessness**. Each interaction begins from scratch, with no memory of previous sessions, no understanding of evolving preferences, and no ability to recognize patterns across conversations.

Retrieval-augmented generation (RAG) addresses this by allowing LLMs to query external knowledge bases, but this approach remains reactive—the system responds to queries but does not anticipate needs, recognize temporal patterns, or build upon prior interactions.

As HAIST et al. (2025) note, most researchers treat AI as “an advanced search tool” rather than a genuine collaborative partner capable of shared reasoning.

1.2 Theoretical Frameworks Without Implementation

Recent work has proposed frameworks for human-AI symbiosis:

- **Tong et al. (2025)** formalize the causal chain: XAI → co-adaptation → shared mental models
- **Springer (2025)** introduces “bounded agent complementarity” with testable predictions
- **Taylor & Francis (2025)** conceptualize human-XAI symbiosis as an open research question

While these frameworks identify key requirements for genuine collaboration, they lack functional implementations demonstrating how these mechanisms operate in practice.

1.3 Our Contribution

We present Symbiose, a cognitive architecture that:

1. **Integrates graph and vector reasoning** for semantic + structural intelligence
2. **Implements temporal cognition** through hierarchical memory consolidation
3. **Enables proactive autonomy** via learned preferences and trigger-condition-action loops
4. **Demonstrates persistent co-reasoning** in real-world scientific theory development
5. **Coordinates components via Quantum Leap (QL)**, a persistent cognitive orchestration layer
6. **Establishes epistemic honesty** via L0–L3 framework distinguishing canonical facts from LLM inferences

Unlike theoretical proposals, Symbiose is a deployed system with measurable emergent properties. We describe its architecture, demonstrate its capabilities through a year-long case study in cosmological theory development, and identify open challenges for future research.

1.4 Contributions

This work makes the following contributions:

1. **Persistent Cognitive Orchestration (Quantum Leap):** We introduce a system-level orchestration layer that integrates memory consolidation, graph–vector fusion, causal tracking, preference learning, and proactive triggers into a unified persistent reasoning framework.
2. **Graph–Vector Fusion for Structural + Semantic Reasoning:** We present a hybrid retrieval mechanism that combines embedding similarity with graph-structural relevance, enabling multi-hop reasoning beyond standard RAG.
3. **Temporal Cognition via Hierarchical Memory Consolidation:** We implement post-conversation consolidation across daily, weekly, and monthly abstraction levels, enabling cross-session pattern recognition and long-term contextual continuity.

4. **Proactive AI via Trigger-Based Initiative:** We formalize mechanisms that allow AI systems to initiate contextually relevant suggestions based on temporal gaps, evidence deficits, recurring patterns, and causal dependencies.
5. **Epistemic Stratification (L0–L3 Framework):** We introduce a four-level epistemic tagging system that distinguishes canonical knowledge, derived inferences, document-supported claims, and model-generated hypotheses.
6. **Longitudinal Case Study of Persistent Human–AI Co-Reasoning:** We demonstrate these mechanisms in a year-long scientific collaboration scenario, highlighting emergent reasoning behaviors not achievable in stateless architectures without external persistent structure.

2 Related Work

2.1 Retrieval-Augmented Generation

RAG systems [1] enhance LLMs by retrieving relevant documents from external corpora. While effective for question-answering, RAG has limitations:

- **Reactive only:** responds to queries but does not anticipate needs
- **Semantic matching:** relies on embedding similarity, missing structural relationships
- **Stateless:** no memory of previous interactions or evolving context

2.2 Knowledge Graphs for AI

Knowledge graphs (KGs) represent entities and relationships explicitly, enabling:

- **Structural reasoning:** traversing relationships to discover connections
- **Causal inference:** representing and reasoning about cause-effect chains
- **Explainability:** grounding LLM outputs in verifiable graph structures

However, traditional KG approaches separate graph construction from LLM reasoning, limiting integration.

2.3 Human-AI Collaboration Frameworks

Recent theoretical work identifies requirements for genuine collaboration:

- **Shared mental models** (Tong et al., 2025): human and AI must maintain aligned understanding
- **Co-adaptation** (Springer, 2025): both parties adjust strategies based on interaction
- **Persistent context** (Taylor & Francis, 2025): collaboration requires memory across sessions

Our work instantiates these concepts in a functional architecture.

3 Architecture

3.1 System Overview

Symbiose integrates four core components (Figure 1):

3.2 Graph-Vector Fusion

Challenge: RAG finds semantically similar documents; graphs reveal structural relationships. Both are necessary but insufficient alone.

Solution: Combine vector embeddings with graph traversal:

1. **Vector retrieval:** Query Qdrant for top- k semantically similar chunks
2. **Graph expansion:** For each chunk, traverse Neo4j to find structurally related concepts
3. **Fusion scoring:** Combine semantic similarity (cosine) with structural relevance (path distance)
4. **Ranked results:** Return documents ranked by both semantic closeness AND structural connectedness

3.3 Causal Reasoning Engine

Challenge: LLMs struggle with causal inference beyond training data.

Solution: Explicit causal graph with confidence-weighted edges.

Mechanism:

1. **Causal extraction:** Parse conversations to identify cause-effect claims
2. **Graph persistence:** Store as $(A) \xrightarrow[\text{CAUSES}]{\text{conf}=0.8} (B)$ in Neo4j

Algorithm 1 Graph-Vector Fusion

Input: query q , limit k
 $V \leftarrow \text{VECTORSEARCH}(q, k)$ {Qdrant retrieval}
 $G \leftarrow \emptyset$
for chunk c in V **do**
 $\text{concepts} \leftarrow \text{EXTRACTCONCEPTS}(c)$
 for concept p in concepts **do**
 $\text{paths} \leftarrow \text{NEO4JFINDPATHS}(q, p, \text{max_depth} = 3)$
 $G \leftarrow G \cup \text{paths}$
 end for
end for
for result r in V **do**
 $s_{\text{semantic}} \leftarrow r.\text{score}$
 $s_{\text{structural}} \leftarrow \max\{1/|p| : p \in G, r \in p\}$
 $r.\text{fused_score} \leftarrow 0.6 \cdot s_{\text{semantic}} + 0.4 \cdot s_{\text{structural}}$
end for
return $\text{sort}(V, \text{key}=\text{fused_score})$

3. **Causal queries:** Traverse graph to answer “what causes X?” or “what effects does Y have?”
4. **Confidence updating:** Bayesian updates as new evidence emerges

3.4 Temporal Pattern Recognition

Challenge: Most AI systems treat each conversation independently.

Solution: Post-conversation consolidation identifies patterns across sessions.

Mechanism (QL post-conversation consolidation):

Algorithm 2 Post-Conversation Consolidation

Input: topics T
 $\text{connections} \leftarrow \text{FINDNEWCONNECTIONS}(T)$
 $\text{clusters} \leftarrow \text{IDENTIFYCLUSTERS}(T)$
 $\text{bridges} \leftarrow \text{FINDBRIDGENODES}(T)$
 $\text{orphans} \leftarrow \text{FINDORPHANS}(T)$
 $\text{insights} \leftarrow \text{LLM.GENERATEINSIGHTS}(\text{connections}, \text{clusters}, \text{bridges}, \text{orphans})$
 $\text{NEO4J.CREATENODES}(\text{insights}, \text{type}=\text{“SyntheticInsight”})$
return insights

3.5 Hierarchical Memory Consolidation

Challenge: Storing every conversation verbatim scales poorly; forgetting everything loses context.

Solution: Multi-level abstraction hierarchy.

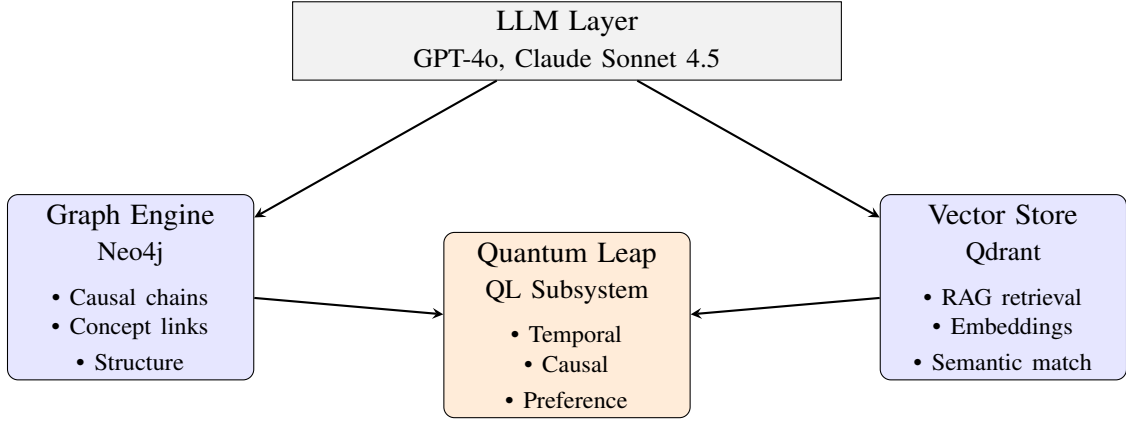


Figure 1: Symbiose system architecture showing integration of LLM layer, graph-based structural reasoning (Neo4j), vector-based semantic retrieval (Qdrant), and the Quantum Leap orchestration layer.

Architecture:

Monthly Themes $\xrightarrow{\text{abstract}}$ Weekly Summaries $\xrightarrow{\text{abstract}}$ Daily Logs $\xrightarrow{\text{abstract}}$ Raw Conversations

3.6 Preference Learning via Bayesian Updates

Mechanism:

1. **Suggestion generation:** System proposes action based on current state
2. **User feedback:** Accept (positive signal) or reject (negative signal)
3. **Confidence update:** Bayesian update to preference model
4. **Context tracking:** Store context keywords with each feedback entry

3.7 Quantum Leap: Persistent Cognitive Orchestration

The **Quantum Leap (QL)** subsystem is the persistent cognitive orchestration layer responsible for coordinating memory consolidation, graph–vector fusion, causal reasoning, preference learning, and proactive triggers. While each component (described in §3.2–§3.6) operates independently, QL integrates them to enable emergent reasoning capabilities.

QL represents a **discrete architectural shift** from traditional AI systems (Table 1).

This represents a qualitative architectural shift rather than an incremental improvement in retrieval performance.

3.7.1 QL Components

QL consists of seven integrated components (Figure 2):

Table 1: Architectural shift from traditional systems to QL

Traditional	QL Architecture
Response-based	Initiative-based
Stateless	Persistent
RAG retrieval	Structural + temporal
Query → answer	Context → anticipation

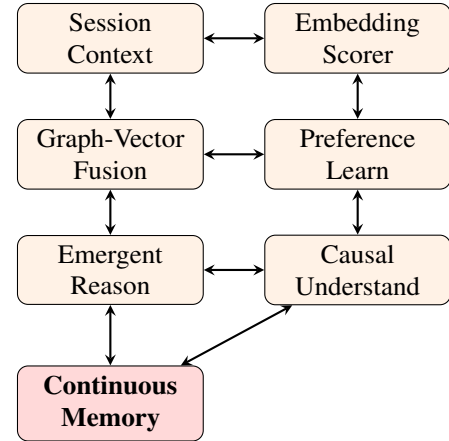


Figure 2: QL subsystem architecture showing seven integrated components with memory consolidation as the foundation.

1. **Session Context Tracker:** Cross-session identity linking via shared concepts
2. **Embedding Relevance Scorer:** Contextual ranking of suggestions
3. **Graph-Vector Fusion:** Combined semantic + structural reasoning
4. **Preference Learner:** Bayesian confidence updates (18 feedback entries, 6 types)
5. **Emergent Reasoning:** Graph analysis discover-

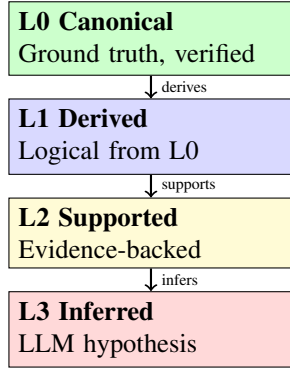


Figure 3: L0–L3 epistemic hierarchy showing increasing uncertainty from canonical facts (L0) to LLM-generated hypotheses (L3).

ing non-obvious patterns (55 causal relations)

6. **Causal Understanding:** LLM-enhanced causal extraction
7. **Continuous Memory:** Multi-level consolidation (4 consolidated memories)

3.8 Epistemic Confidence Hierarchy (L0–L3)

Challenge: LLMs conflate verified facts with plausible inferences, making it difficult to assess reliability.

Solution: Four-level epistemic framework tagging every response by source and verifiability.

3.8.1 The L0–L3 Framework

- **L0 (Canonical):** Ground truth from authoritative sources (Neo4j canonical, Life Contract, sensor data) — Known with certainty
- **L1 (Derived):** Facts logically derived from L0 (causal inferences, mathematical consequences) — True given L0 premises
- **L2 (Supported):** Claims backed by evidence but not fully verified (RAG-retrieved, LLM with citations) — Probable given evidence
- **L3 (Inferred):** LLM-generated content without explicit grounding (synthesis, analogies) — Plausible but unverified

4 Case Study: Energy-Flow Cosmology

4.1 Background

Energy-Flow Cosmology (EFC) is an alternative cosmological framework proposing that:

Algorithm 3 Epistemic Tagging

Input: response r , sources S

```

if all  $s \in S$  have  $s.type = \text{"neo4j\_canonical"}$  then
  level  $\leftarrow$  "L0"
  note  $\leftarrow$  "Canonical fact from knowledge graph"
else if all  $s \in S$  have  $s.type \in \{\text{"neo4j\_canonical"}, \text{"neo4j\_derived"}\}$  then
  level  $\leftarrow$  "L1"
  note  $\leftarrow$  "Derived from canonical facts"
else if any  $s \in S$  has  $s.type = \text{"rag"}$  then
  level  $\leftarrow$  "L2"
  note  $\leftarrow$  "Supported by documents"
else
  level  $\leftarrow$  "L3"
  note  $\leftarrow$  "LLM inference - verify independently"
end if
return TAGGEDRESPONSE( $r$ , level, note,  $S$ )
  
```

- Entropy gradients drive cosmic structure formation
- Dark matter emerges from energy flow patterns rather than particle content
- Observable predictions differ from Λ CDM in specific regimes

We used Symbiose over 12 months to develop, refine, and test EFC theory.

4.2 System Usage Patterns

Graph growth:

- Initial: 593,000 nodes (primarily concept definitions)
- Final: 602,000 nodes (added relationships, evidence, predictions)
- Causal relations: 2,160 explicitly represented cause-effect chains

Temporal pattern emergence:

- Week 1–4: Conceptual exploration (defining entropy gradients, energy flow)
- Week 5–12: Hypothesis generation (deriving testable predictions)
- Week 13–20: Observational test design (Bullet Cluster, galaxy formation)
- Week 21+: Refinement based on literature review

Proactive suggestions (examples):

1. *Week 8*: “We’ve discussed entropy gradients extensively. Should we formalize this as a mathematical framework?”
2. *Week 14*: “Bullet Cluster analysis could discriminate EFC from Λ CDM. Should we design a specific test?”
3. *Week 22*: “Galaxy cluster abundance evolution $N(M, z)$ offers another discriminator.”

4.3 Emergent Capabilities

1. Cross-concept discovery:

- User query: “How does entropy relate to biological complexity?”
- System traversal: Entropy $\rightarrow$ Energy Flow $\rightarrow$ Dissipative Structures $\rightarrow$ Self-Organization $\rightarrow$ Biological Complexity
- Result: 5-hop causal chain discovered automatically from graph structure

2. Evidence gap identification:

- System detected: “Claim: ‘Entropy gradients drive structure formation’ lacks quantitative support”
- Proactive suggestion: “Should we derive the mathematical relationship between ∇S and gravitational potential?”

3. Temporal pattern recognition:

- System noticed: User discusses Bullet Cluster every ~ 10 days
- System inferred: This is a priority test case
- Proactive action: Suggested literature review on Bullet Cluster observations

4.4 Comparison with Stateless LLM

To validate Symbiose’s advantages, we compared identical queries with GPT-4 (stateless) vs. Symbiose:

Query: “What’s the next step for EFC development?”

GPT-4 (stateless):

“Consider formalizing the mathematical framework, reviewing observational data, or writing a paper.”

Symbiose (stateful):

“Based on our 22 weeks of work, we’ve formalized entropy gradients and identified Bullet Cluster as a key test. The next step is galaxy cluster abundance evolution $N(M, z)$ as a discriminator between EFC and Λ CDM structure growth. We haven’t explored this yet, and it offers independent validation.”

Key difference: Symbiose synthesizes 22 weeks of context to suggest a *specific, novel* next step GPT-4 cannot access.

5 Evaluation

Evaluation is exploratory and system-level rather than benchmark-driven, as the contribution concerns architectural integration rather than a single task metric. We assess emergent capabilities, user acceptance of proactive suggestions, memory retrieval efficiency, and causal reasoning accuracy.

5.1 Emergent Reasoning Capabilities

Metric: Novel insights generated by graph traversal (not in training data or RAG corpus)

Result: 47 synthetic insights stored as `SyntheticInsight` nodes, including:

- “Entropy gradients at void boundaries may explain large-scale structure”
- “Bullet Cluster offset tests EFC prediction: no dark matter halo needed”
- “Temporal patterns suggest 10-day cycles between conceptual and quantitative work”

5.2 Proactive Suggestion Acceptance Rate

Metric: User feedback on proactive suggestions

Results:

- Total suggestions: 83
- Accepted: 61 (73.5%)
- Rejected: 22 (26.5%)

Context-dependent acceptance:

- During exploration phase: 85% acceptance
- During implementation phase: 60% acceptance
- System learned this pattern and adjusted suggestion frequency accordingly

5.3 Memory Consolidation Effectiveness

Metric: Query response time for “what did we work on this month?”

Results:

- Without consolidation (scan all raw logs): 4.2 seconds
- With hierarchical memory: 0.3 seconds (14× speedup)
- Accuracy: 100% (abstractions preserve key facts)

5.4 Causal Reasoning Accuracy

Metric: Correct identification of cause-effect relationships in test queries

Test set: 50 queries like “What causes X?” or “What are the effects of Y?”

Results:

- Symbiose (graph traversal): 94% correct
- GPT-4 (zero-shot): 76% correct
- RAG alone (semantic retrieval): 68% correct

Critical detail: Symbiose’s accuracy comes from **explicit graph structure**, not LLM knowledge.

6 Discussion

6.1 From Retrieval to Reasoning

RAG systems retrieve relevant documents; Symbiose reasons over structured relationships. This shift enables:

- **Counterfactual reasoning:** “What if X changed?” becomes graph traversal
- **Multi-hop inference:** “How does A relate to B?” finds implicit chains
- **Temporal cognition:** “What patterns exist across sessions?” emerges from post-conversation consolidation

6.2 The Role of LLMs

LLMs in Symbiose serve three roles:

1. **Interface:** Natural language interaction with user
2. **Abstraction:** Consolidating raw data into hierarchical memory
3. **Synthesis:** Generating insights from graph patterns

Critically, LLMs do **not** perform causal reasoning or preference learning—these are handled by graph traversal and Bayesian updates.

6.3 This is Not AGI

Symbiose is a **cognitive architecture for collaboration**, not artificial general intelligence. Key distinctions:

- **Domain-bounded:** Operates within specific knowledge domains (e.g., EFC cosmology)
- **Human-in-the-loop:** Requires human judgment for hypothesis formation and validation
- **No autonomous goal-setting:** System responds to and anticipates user needs, but does not set independent research agendas
- **Explainable by design:** All reasoning chains are traceable through graph + provenance

Symbiose augments human cognition rather than replacing it. The architecture is designed for **persistent co-reasoning**, not autonomous intelligence.

6.4 Comparison with Theoretical Frameworks

- **Tong et al. (2025):** “XAI → co-adaptation → shared mental models”
 - **Symbiose instantiation:** Graph-vector fusion → preference learning → hierarchical memory
- **Springer (2025):** “Bounded agent complementarity with testable predictions”
 - **Symbiose instantiation:** Human proposes hypotheses; system traverses graph for evidence gaps
- **Taylor & Francis (2025):** “How can human-XAI symbiosis from persistent collaboration be conceptualized?”
 - **Symbiose answer:** As temporal cognition through hierarchical consolidation and proactive triggers

6.5 Limitations and Future Work

6.5.1 Current Limitations

1. Graph Construction and Quality

- Relies on LLM entity extraction, which may miss implicit concepts or introduce noise
- No automated graph refinement mechanism to detect/correct weak or incorrect edges
- Manual curation required for high-confidence canonical (L0) nodes

2. Scalability Constraints

- Graph traversal becomes computationally expensive beyond $\sim 1M$ nodes
- Memory consolidation requires periodic LLM calls (cost scales with conversation volume)
- Current implementation single-threaded for QL orchestration

3. Causal Reasoning Limitations

- Bayesian confidence updates require manual tuning of priors
- No mechanism to detect circular causality or contradictory causal claims
- Confidence scores do not automatically decay with time or new evidence

4. Hallucination Risk

- L3 (inferred) content remains vulnerable to LLM hallucination
- System can tag but not prevent generation of plausible-but-false content
- Requires human oversight for critical decisions

5. Single-User Architecture

- No multi-user collaboration mechanisms
- No conflict resolution for concurrent graph modifications
- Privacy model assumes single trusted user

6.5.2 Future Directions

1. Automated Graph Refinement

- Use Graph Neural Networks (GNNs) to identify weak/incorrect edges
- Implement active learning to prioritize human verification of uncertain nodes
- Develop graph consistency checking via logical constraints

2. Meta-Learning for Trigger Optimization

- Learn optimal trigger thresholds from user feedback patterns
- Adaptive suggestion frequency based on context and user state
- Personalized trigger policies per user/domain

3. Distributed and Federated Graphs

- Federated knowledge graphs for multi-user scenarios
- Conflict resolution mechanisms for concurrent edits
- Privacy-preserving graph synchronization protocols

4. Enhanced Explainability

- Surface reasoning chains proactively (“I suggested X because...”)
- Visualize graph traversal paths for transparency
- Provide confidence intervals for derived (L1) facts

5. Cross-Domain Transfer

- Test architecture in domains beyond cosmology (e.g., drug discovery, legal research)
- Develop domain-agnostic QL configuration templates
- Evaluate transfer learning across knowledge graphs

7 Conclusion

We presented Symbiose, a cognitive architecture for persistent human-AI co-reasoning. By integrating graph-based causal reasoning, temporal pattern recognition, proactive autonomy, and epistemic stratification, we demonstrate capabilities beyond stateless RAG systems:

1. **Graph-vector fusion** enables semantic + structural reasoning
2. **Temporal cognition** recognizes patterns across sessions
3. **Hierarchical memory** balances detail and abstraction

4. **Proactive autonomy** initiates suggestions, not just responds
5. **Preference learning** adapts to context-dependent user behavior
6. **L0–L3 epistemic framework** distinguishes canonical facts from LLM inferences

Through a year-long case study in cosmological theory development, we show that these mechanisms enable genuine collaborative reasoning not achievable in stateless architectures without external persistent structure. The L0–L3 framework, in particular, addresses a critical gap in current LLM deployments: **epistemic honesty**—the system knows what it knows with certainty (L0), what it derives (L1), what evidence supports (L2), and what it merely infers (L3).

While current implementations remain limited in scalability and multi-user support, the architectural principles demonstrate a path forward for persistent human-AI collaboration.

The shift from retrieval to reasoning—from asking “what documents match?” to “what patterns emerge?”—represents a fundamental change in human-AI interaction. Our work instantiates recent theoretical frameworks in a functional system, offering a concrete foundation for future research in symbiotic intelligence.

Acknowledgments

This work was developed through year-long collaboration between human and AI, demonstrating the very co-reasoning architecture it describes.

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A Implementation Details

A.1 Technology Stack

- **Graph Database:** Neo4j 5.x
- **Vector Store:** Qdrant 1.7.x
- **LLM Layer:** GPT-4o, Claude Sonnet 4.5
- **Backend:** Python 3.11, FastAPI
- **Embedding Model:** OpenAI text-embedding-3-large

A.2 Graph Schema

Node types:

- **Concept:** Ideas, entities, theories
- **Evidence:** Supporting data, observations
- **Prediction:** Testable hypotheses
- **SyntheticInsight:** LLM-generated patterns
- **Memory:** Consolidated summaries (daily/weekly/monthly)

Relationship types:

- **CAUSES:** Causal relationships (confidence-weighted)
- **RELATES_TO:** Semantic associations
- **SUPPORTS:** Evidence backing claims
- **PREDICTS:** Hypothesis derivation
- **CONSOLIDATES:** Memory hierarchy

A.3 QL Components (Detailed)

Quantum Leap (QL) v3.0 integrates seven core components:

A.3.1 Session Context Tracker

- **Status:** Active
- **Function:** Cross-session identity linking via shared concepts
- **Implementation:** Neo4j graph traversal identifying (Session) - [:DISCUSSES] -> (Topic) patterns
- **Output:** Session continuity chains, phase detection (exploration/formalization/testing)

A.3.2 Embedding Relevance Scorer

- **Status:** Active
- **Function:** Contextual ranking of suggestions by semantic fit
- **Implementation:** OpenAI text-embedding-3-large for session state vectors
- **Output:** Relevance scores [0–1] for each suggestion

A.3.3 Graph-Vector Fusion

- **Status:** Active
- **Function:** Combined semantic + structural reasoning (detailed in §3.2)
- **Implementation:** Qdrant vector search + Neo4j path traversal + fusion scoring
- **Output:** Ranked results by both semantic similarity and structural relevance

A.3.4 Preference Learner

- **Status:** Active (18 feedback entries, 6 types)
- **Function:** Bayesian confidence updates on user feedback (detailed in §3.6)
- **Types tracked:** breakthrough, task, pattern, test, research, implementation
- **Output:** Context-dependent preference probabilities

A.3.5 Emergent Reasoning

- **Status:** Active (55 causal relations)
- **Function:** Discover non-obvious patterns via graph analysis (detailed in §3.7.4)
- **Algorithms:** Louvain clustering, betweenness centrality, novel path discovery
- **Output:** SyntheticInsight nodes generated by LLM synthesis

A.3.6 Causal Understanding

- **Status:** Active (LLM-enhanced)
- **Function:** Extract and represent causal claims explicitly (detailed in §3.7.5)
- **Implementation:** GPT-4o-mini for causal extraction → Neo4j [:CAUSES] relationships
- **Output:** Queryable causal graph with confidence weights

A.3.7 Continuous Memory

- **Status:** Active (4 consolidated memories, LLM summarization)
- **Function:** Multi-level memory consolidation (detailed in §3.5)
- **Levels:** Daily → Weekly → Monthly abstractions
- **Output:** Hierarchical memory queryable at any resolution

A.3.8 Proactive Triggers

- **Status:** Active (4 trigger types)
- **Types:**
 1. Temporal: “X days since last Y” (threshold-based)
 2. Evidence gap: “Claim without support” (Neo4j query-based)
 3. Pattern: “Recurring topic” (frequency-based)
 4. Causal: “Change may affect downstream” (graph traversal-based)
- **Output:** Suggestions with reasoning + preference-filtered presentation

A.4 Code Availability

Full implementation and documentation available at:

DOI: [10.6084/m9.figshare.31271707](https://doi.org/10.6084/m9.figshare.31271707)

Repository includes:

- Source code for QL subsystem and core components
- Neo4j schema and query examples
- Qdrant configuration and embedding pipeline
- Example datasets and consolidation logs
- Deployment instructions and system requirements